Question	Answer	Marks	Partial Marks
3(a)(ii)	51 200 or 51 190 or 51 194	**2**	**M1** for $\frac{35.8}{100} \times 143\,000$ After 0 scored SC1 for answer figs 512 or figs 5119 or figs 51194
3(a)(iii)	2.43 or 2.434…	**1**	
3(a)(iv)	3000 or 3004 to 3005	**2**	**M1** for $\frac{4.495\times10^9}{1.496\times10^8}$ [× 100] oe After 0 scored SC1 for answer figs 3 or figs 3004…. or figs 3005
3(a)(v)	1.52 or 1.522…	**2**	**B1** for 1AU = 1.5[0] × 10^8 or 1.497… × 10^8 [km] or $1\text{km} = 6.68\times10^{-9}$ *or* $6.678\ldots\times10^{-9}$ [AU] OR **M1** for $\frac{5.2\times2.279[\times10^8]}{7.786[\times10^8]}$ oe After 0 scored **SC1** for answer figs 152 or figs 1522……
3(a)(vi)	4890 or 4885…	**2**	**M1** for $d \times \left(1+\frac{39.2}{100}\right)$ = 6800 oe
3(b)(i)	2.9979 × 10^5 × 60^2 × 24 × 365.25	**M1**	After **M0** **SC1** for 2.9979 × 10^5 × 31557600 oe
	= 9.4606… × 10^{12}	**A1**	
3(b)(ii)	2.54 or 2.536 to 2.537	**2**	**M1** for $\frac{2.4\times10^{19}}{9.461\times10^{12}}$ oe
4(a)(i)	$\frac{2}{5}$ $\frac{5}{9}$, $\frac{4}{9}$ $\frac{5}{9}$, $\frac{4}{9}$	**2**	**B1** for $\frac{2}{5}$ and a pair of probabilities for spinner B that sum to 1
4a(ii)(a)	$\frac{1}{3}$ oe	**2**	**FT dep** *their* tree diagram **M1** for $\frac{3}{5}$ × *their* $\frac{5}{9}$
4a(ii)(b)	$\frac{2}{3}$ oe	**1**	**FT** dep 1 – *their* $\frac{1}{3}$
4(b)	72	**1**	

Question	Answer	Marks	Partial Marks
4(c)	$\frac{20}{81}$ oe	**3**	**M2** for $\frac{2}{9}\times\frac{4}{9}$ [× 2] + $\frac{2}{9}\times\frac{2}{9}$ oe or **M1** for $\frac{2}{9}\times\frac{4}{9}$ or $\frac{2}{9}\times\frac{2}{9}$ oe
4(d)	$\left(\frac{5}{9}\right)^{n-1}$ [×]$\frac{4}{9}$ oe final answer	**2**	**M1** for $\left(\frac{5}{9}\right)^{n-1}$ seen
5(a)	27.3 or 27.32 to 27.33	**5**	**M4** for tan[$\angle ACD$] = $\frac{83.2}{\frac{83.2}{\tan 38}+54.5}$ oe or **M3** for [AC =] $\frac{83.2}{\tan 38}$ +54.5 oe or for [CD =] $\sqrt{54.5^2+\left(\frac{83.2}{\sin 38}\right)^2-2(54.5)\left(\frac{83.2}{\sin 38}\right)\cos(180-38)}$ oe or **M2** for [AB =] $\frac{83.2}{\tan 38}$ oe or for [BD =] $\frac{83.2}{sin38}$ oe or **M1** for tan38 = $\frac{83.2}{AB}$ oe or sin38 = $\frac{83.2}{BD}$ oe
5(b)	Centre marked at midpoint of *FG*. and **Angle** in a **semi-circle** is **90**	**B2**	**B1** for marking the centre at mid-point of FG